Md. Almabud **Juwal**

Senior Software Engineer | Backend, Data & Architecture

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Dhaka, Bangladesh

Summary

Senior Software Engineer with more than six years of experience building backend and data-intensive systems in Python, FastAPI, Django, and PostgreSQL. Experienced in system architecture, performance optimization, asynchronous data pipelines, and leading small engineering teams. Recent work includes a machine-learning pipeline for entity matching and an MCP server for language-model report generation.

Skills

Programming	Python, SQL, JavaScript/TypeScript, Go, C/C++
Backend & Frameworks	FastAPI, Django, Django REST Framework, Celery
Data & Databases	PostgreSQL, Pandas, Redis, MongoDB, Neo4j, Elasticsearch
Machine Learning & AI	scikit-learn, Sentence Transformers, RapidFuzz, Azure Document Intelligence, MCP
Infrastructure & Engineering	Docker, RabbitMQ, Git, CI/CD, Jenkins, Ansible
Frontend	Angular

Experience

Present – Jun 2024 — Senior Software Engineer | Cogwheel Analytics, Colorado, United States (Remote)

- Architect the next-generation **Cogwheel Analytics** platform as a **modular monolith** following **Clean Architecture** principles, with reusable handling for authentication, pagination, and request validation that reduces repetitive implementation work
- Lead the development and maintenance of the **Data Cleaning** service, implementing data-validation and transformation workflows to improve data integrity and consistency
- Designed and implemented a hotel-name matching pipeline that combines TF-IDF and fuzzy-string retrieval, sentence-embedding search, and cross-encoder re-ranking with threshold-based acceptance
- Built an MCP server that lets a language model generate custom reports from analytics data, and worked on extracting data from PDF reports with Azure Document Intelligence
- Design and implement backend services, APIs, and asynchronous data-processing and web-scraping workflows

Python | FastAPI | PostgreSQL | Pandas | Celery | Scrapy | scikit-learn | Sentence Transformers | MCP

May 2024 – Jul 2023 — Senior Software Engineer | Brain Station 23 Ltd, Dhaka, Bangladesh

- Led the turnaround of a delayed project by mentoring **3 junior engineers**, restructuring task allocation, and redesigning the system architecture, enabling successful delivery of the second version within **6 months**
- Designed software architectures for **Task Station** and **Sayvery**, with focus on scalability, maintainability, and clear separation of responsibilities
- Managed the technical backlog by translating product and stakeholder requirements into prioritized engineering tasks aligned with project goals and delivery timelines
- Designed and implemented web-scraping and invoice-parsing solutions to automate data collection and extraction

Python | Django | Django REST Framework | PostgreSQL | Celery | Scrapy

Jun 2023 – Oct 2021 — Software Engineer | Brain Station 23 Ltd, Dhaka, Bangladesh

- Improved system performance by approximately **10%** through database-query optimization and API response time by approximately **5%** through application-level caching
- Upgraded Python and Django versions and refactored the existing codebase
- Contributed to real-time synchronization between backend and frontend systems using WebSockets and Go, while mentoring an Associate Software Engineer

Python | Django | Django REST Framework | PostgreSQL | Redis | Celery | Go

Sep 2021 – Jul 2020 | **Junior Software Engineer | Tirzok Private Limited, Dhaka, Bangladesh**

- Reduced processing time for **3.3 million records** from approximately **29–35 minutes to 2–4 minutes** using PostgreSQL COPY and database triggers
- Reduced database query response time by approximately **50%** through query optimization
- Contributed to backend and database architecture, including the implementation of a custom authentication backend and graph-based data solutions using Neo4j
- Automated deployment workflows using Jenkins and Ansible

Python · Django · PostgreSQL · Redis · Neo4j · Jenkins · Ansible · Angular

Apr 2020 – Dec 2019 | **Web Developer | Wizdoor Inc, Dhaka, Bangladesh**

- Contributed to backend and database architecture while coordinating implementation with frontend and design teams
- Developed a real-time chat application using Django Channels and WebSockets

Python · Django · Django REST Framework · PostgreSQL · Django Channels · Angular

Projects

Rehive Oct 2021 – Feb 2023

https://rehive.com/

Contributed to a South Africa-based fintech platform supporting multiple currencies, including cryptocurrency. Implemented the billing service and contributed to a plugin-based architecture that enables additional services and commerce-oriented functionality to be integrated with the platform.

Python · Django · Django REST Framework · PostgreSQL · Celery · RabbitMQ · Redis · Docker · Angular

LEGISLIB May 2020 – Jul 2020

Legal-document processing and search platform for case-law materials from Bangladesh, India, the UK, and the USA. Processed and reformatted PDF documents and indexed legal content to provide searchable access for Bangladeshi legal professionals and students.

Python · Django · Django REST Framework · MongoDB · Elasticsearch · PDF Processing

Education

2020 – 2016 | **B.Sc. in Computer Science and Engineering**

- **Daffodil International University**, Dhaka, Bangladesh
- **CGPA : 3.77 / 4.00**
- **Final-year project (A+) :** mobile application that reads exam marks from photographed cover pages with the Microsoft Azure Vision API

Certifications & Courses

Jun 2025 Advanced Certificate for Management Professionals (ACMP 4.0), University of Dhaka
Apr 2023 AWS Certified Cloud Practitioner
2021 Object-Oriented Design, Coursera (course)
2021 The Ultimate Docker Course, Mosh Hamedani (course)
Sep 2020 HackerRank Python (Basic) Certificate

Academic & Programming Achievements

2018 Participant, ACM ICPC Dhaka Preliminary Contest
2018 **4th place**, DIU Intra Team Programming Contest
2017 **2nd place**, DIU Intra Team Programming Contest
2017 **9th place**, DIU Takeoff Individual Programming Contest

Languages

Bengali : Native **English** : IELTS Academic 6.5 (CEFR B2) **Hindi** : Conversational